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FINAL YEAR PROJECT REPORT

AI-Powered Code Review and Bug Prediction System

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A Final Year Project Report submitted in partial fulfillment of the requirements for the Degree of BCS (Hons.) in Computer Science

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CERTIFICATE OF APPROVAL

This is to certify that the Final Year Project titled:

"AI-Powered Code Review and Bug Prediction System"

submitted by Muhammad Nasir Khan, Ahmed Khan Khisro, and Umair Dost has been completed under my supervision and meets the criteria for approval as a Final Year Project Report in partial fulfillment of the requirements for the Degree of BCS (Hons.) at the Institute of Management Sciences, Peshawar. In my opinion, this work satisfies the required standards of originality, quality, and technical depth expected at this level.

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DECLARATION OF ORIGINALITY

We, Muhammad Nasir Khan, Ahmed Khan Khisro, and Umair Dost, hereby solemnly declare that the Final Year Project Report entitled "AI-Powered Code Review and Bug Prediction System" submitted to the FYP Coordinator and Research & Development Directorate (R&DD) of the Institute of Management Sciences, Peshawar, is our own original work.

We further declare that:

- This report has not been submitted previously, in full or in part, for any degree or examination at this or any other institution.
- The work presented herein is entirely our own and all sources of information have been duly acknowledged and cited in accordance with standard academic practice.
- We are aware that any plagiarism or academic dishonesty, if discovered, will result in immediate cancellation of the project and may subject us to further disciplinary action as per the academic regulations of IMSciences.
- All experiments, implementations, and evaluations described in this report were conducted by the authors unless explicitly stated otherwise.

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BCS (Hons.) 2021–2025	BCS (Hons.) 2021–2025	BCS (Hons.) 2021–2025

DEDICATION

This work is dedicated to

our parents and families,

whose unconditional love, prayers, and sacrifices
have been the foundation of everything we have achieved.

And to our teachers and mentors,
who lit the path with knowledge and guided us with patience.

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Finally, we owe an immeasurable debt of gratitude to our families. Their prayers, patience, emotional support, and financial sacrifices over the course of our studies have been our greatest source of strength. This achievement belongs as much to them as it does to us.

Muhammad Nasir Khan • Ahmed Khan Khisro • Umair Dost

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ABSTRACT

Software defects represent a persistent and economically significant challenge in modern software development. The manual review of source code, while effective when performed by experienced engineers, is resource-intensive, inherently inconsistent, and does not scale adequately with the growing complexity and pace of contemporary software projects. Studies have consistently shown that defects discovered in post-release stages cost up to fifteen times more to remediate than those identified during the development phase, making early and automated defect detection a high-value research and engineering objective.

This project presents an AI-Powered Code Review and Bug Prediction System — a comprehensive, data-driven solution that leverages state-of-the-art machine learning techniques to automate two complementary code quality tasks: the detection of code smells (structural quality violations) and the prediction of bug-prone software modules. The system employs a novel hybrid approach that combines the complementary representational strengths of Long Short-Term Memory (LSTM) recurrent neural networks, operating on tokenised raw source code sequences, with Random Forest classifiers trained on a rich set of 87 engineered software metrics extracted through static analysis.

Training and evaluation data were collected by systematically mining 512 open-source GitHub repositories in Python and Java, yielding 18,742 labelled code modules. Bug-introducing commits were identified using the SZZ algorithm applied to commit message metadata. Eighty-seven software metrics spanning size, complexity, object-oriented structure, process history, and inter-module dependency dimensions were computed per module. After feature selection (removing highly correlated and near-zero variance features), 61 features were retained for the Random Forest model. Class imbalance was addressed using SMOTE oversampling.

The hybrid ensemble model, combining Random Forest (weight: 0.55) and LSTM (weight: 0.45) via soft voting, achieved an accuracy of 87.1%, precision of 0.84, recall of 0.89, F1-score of 0.87, and AUC-ROC of 0.94 on a held-out test set of 2,812 modules. These results represent consistent improvements over all evaluated baselines, including SonarQube (+15.9% accuracy), PMD (+18.6%), Logistic Regression (+10.8%), and Naive Bayes (+13.0%). SHAP (SHapley Additive exPlanations) values are used to produce metric-level explanations for each prediction, generating actionable, human-readable review suggestions categorised by severity.

The complete system is deployed through a FastAPI REST backend exposing prediction endpoints, integrated with an Angular 16 web dashboard that provides a code submission interface, severity-colour-coded annotation display, and downloadable review report generation in JSON and PDF formats.

Keywords: *code smell detection, bug prediction, machine learning, LSTM, Random Forest, SHAP explainability, static code analysis, software metrics, GitHub mining, Mining Software Repositories, automated code review, FastAPI, Angular.*

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
LSTM	Long Short-Term Memory
RF	Random Forest
NLP	Natural Language Processing
API	Application Programming Interface
REST	Representational State Transfer
AST	Abstract Syntax Tree
CFG	Control Flow Graph
PDG	Program Dependence Graph
SZZ	Śliwerski–Zimmermann–Zeller (bug-finding algorithm)
MSR	Mining Software Repositories
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
CBO	Coupling Between Object Classes
WMC	Weighted Methods per Class
RFC	Response for a Class
LCOM	Lack of Cohesion in Methods
DIT	Depth of Inheritance Tree
NOC	Number of Children
LOC	Lines of Code
LLOC	Logical Lines of Code
FYP	Final Year Project
BCS	Bachelor of Computer Science
IMSciences	Institute of Management Sciences

HHN	Hochschule Heilbronn
JSON	JavaScript Object Notation
PDF	Portable Document Format
PMD	Programming Mistake Detector
IDE	Integrated Development Environment
VS Code	Visual Studio Code
GPU	Graphics Processing Unit
CPU	Central Processing Unit
RAM	Random Access Memory

CHAPTER 1: INTRODUCTION

1.1 Background and Motivation

The proliferation of large-scale, interconnected software systems has intensified the importance of rigorous code quality assurance. Modern software applications underpin critical infrastructure — from financial transaction platforms and healthcare management systems to autonomous vehicles and cloud-based communication services. In this landscape, software defects are not merely an inconvenience; they represent a direct risk to reliability, security, and organisational reputation.

Empirical research consistently demonstrates the economic significance of software defects. Capers Jones' comprehensive studies of software development economics indicate that defect removal consumes between 40% and 50% of total development effort across the industry. More specifically, defects discovered during post-release maintenance phases cost between ten and fifteen times more to remediate than those identified during active development, and up to forty to sixty times more than defects caught during the requirements phase. These statistics underscore a fundamental challenge: the earlier a defect is detected, the cheaper and more tractable its resolution.

Manual code review remains the dominant quality assurance practice in professional software development. Studies by Bacchelli and Bird (2013) at Microsoft Research identified defect detection, code improvement, and knowledge transfer as the primary motivations for code review. However, manual review is constrained by inherent human limitations: reviewers experience cognitive fatigue, harbour unconscious biases toward familiar coding patterns, and operate under time pressure in Agile development environments with rapid release cycles. As codebases grow to millions of lines of code across hundreds of interconnected modules, the adequacy of manual review as a sole quality gate becomes untenable.

Static analysis tools such as SonarQube, PMD, FindBugs, and Checkstyle have emerged to augment manual review, applying predefined rule sets to identify patterns associated with defects and poor code quality. While these tools provide a degree of automation and consistency, they suffer from well-documented limitations: their rule-based nature produces false positive rates often exceeding 60%, leading to alert fatigue and developer distrust; they cannot learn from project-specific historical defect data; and they fail to capture the emergent, context-dependent patterns that characterise real-world bug-prone code.

The rapid maturation of machine learning techniques — particularly deep learning architectures capable of learning rich representations from raw source code — has opened compelling avenues for intelligent, adaptive code quality assessment. The ability to predict

which modules in a codebase are most likely to contain defects, prior to manual inspection and prior to testing, can substantially reduce the search space for reviewers, enable more targeted testing resource allocation, and ultimately improve software reliability at scale.

This project is motivated by three converging observations: (1) the limitations of existing automated tools are well-established and leave a clear performance gap; (2) the AI and ML research community has demonstrated that both structured metric-based and sequence-based deep learning approaches capture predictive signals that rule-based systems cannot encode; and (3) the practical utility of published research systems is limited by their lack of accessible interfaces and actionable output formats. This project addresses all three gaps simultaneously.

1.2 Problem Statement

The central problem addressed by this project can be stated as follows: existing automated code review tools fail to provide accurate, adaptive, and actionable defect predictions because they rely on static rule sets that do not learn from historical defect data and do not capture the complex, contextual patterns that distinguish bug-prone code from clean code.

More specifically, four problems are identified:

- High false positive rates in rule-based tools: SonarQube and PMD report false positive rates consistently above 60% in industrial evaluations, leading developers to disable or ignore automated alerts — defeating their purpose.
- Inability to adapt to project-specific patterns: rule-based tools apply universal rule sets that cannot account for domain-specific coding conventions, team-specific defect patterns, or the evolving quality characteristics of individual codebases.
- Isolation of approaches in published research: the majority of academic defect prediction systems exploit either structured metric-based features or deep learning on raw code, but not both. The complementary strengths of these two modalities remain underutilised.
- Lack of accessible, actionable interfaces: even high-performing research systems rarely expose their functionality through usable APIs or dashboards, limiting their adoption beyond the research community.

1.3 Objectives

The primary objectives of this project are:

1. To design and implement a machine learning pipeline for bug-prone module prediction using 87 software metrics extracted from real-world GitHub repositories through static analysis.
2. To develop and train an LSTM-based deep learning model for sequential source code pattern recognition, capable of detecting code smells through learned representations of tokenised code.
3. To construct a hybrid ensemble system that combines the complementary predictive strengths of the LSTM and Random Forest models via soft voting, achieving superior performance to either model alone.
4. To implement a production-quality RESTful API backend using FastAPI and an Angular-based web dashboard providing a complete, end-to-end code review interface.

5. To evaluate the system rigorously against established baselines (SonarQube, PMD, Logistic Regression, Naive Bayes) on a standardised test set, using accuracy, precision, recall, F1-score, and AUC-ROC as evaluation metrics.
6. To implement model explainability through SHAP (SHapley Additive exPlanations) values, enabling the system to generate actionable, metric-level review suggestions ranked by feature contribution.

1.4 Scope of the Project

This project encompasses the full pipeline from raw data collection to system deployment. Specifically, the scope includes: automated mining of GitHub repositories using the GitHub REST API; static metric extraction using Radon (Python) and CKJMExtended (Java); dataset construction, preprocessing, and class balancing using SMOTE; model development, hyperparameter tuning, and ensemble integration; REST API development and Angular frontend implementation; and rigorous empirical evaluation against baselines.

The current implementation is scoped to Python and Java codebases. The system analyses entire code modules (files) rather than individual commits or code diffs. Automated code repair, version control hook integration, and IDE plugin development are identified as areas for future work and are outside the current scope.

1.5 Significance of the Work

This work makes the following contributions to the field of automated software quality assurance:

- A novel hybrid ML/DL architecture that jointly exploits structured software metrics and raw source code sequences, demonstrating that ensemble integration of these complementary modalities yields performance superior to either approach in isolation.
- A large-scale empirical evaluation on a dataset of 18,742 labelled modules mined from 512 GitHub repositories, providing a reproducible benchmark for future comparison.
- A fully deployed, API-accessible system with an Angular dashboard and SHAP-based explanations, bridging the gap between research performance and practical developer utility.
- A structured, severity-ranked review suggestion format that translates raw model predictions into actionable developer guidance, addressing the usability shortcomings of existing research tools.

1.6 Report Structure

The remainder of this report is organised as follows. Chapter 2 presents a comprehensive review of the relevant literature, covering traditional code review, machine learning for software engineering, bug prediction models, and deep learning for code analysis. Chapter 3 describes the methodology in detail, including system architecture, data collection, feature engineering, model development, and system implementation. Chapter 4 presents the experimental setup, performance results, baseline comparisons, and qualitative system output

analysis. Chapter 5 concludes the report with a discussion of limitations and future research directions. References and appendices follow.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

This chapter surveys the body of research relevant to the central themes of this project: the evolution of code review practices, the application of machine learning to software engineering problems, the development of defect prediction models, and the emerging role of deep learning in source code analysis. The chapter concludes by identifying the research gaps that motivated the design choices of this project.

2.2 Traditional Code Review Approaches

Code review has been a cornerstone of software quality assurance since the introduction of Fagan Inspections in the 1970s. Michael Fagan's landmark 1976 paper at IBM introduced the concept of structured code inspections as a formal process for defect detection, demonstrating that early defect removal through peer inspection was both more effective and more economical than later-stage testing. Fagan's work established the empirical foundation for code review as a primary quality gate and influenced decades of subsequent practice.

The operationalisation of code review has evolved considerably since Fagan's formal inspections. Wiegers (2002) documented a spectrum of review practices from formal inspections through walkthroughs to lightweight peer reviews, noting that the cost-benefit profile of each approach depends heavily on team size, project risk, and time constraints. The emergence of distributed version control systems and pull request workflows — popularised by platforms such as GitHub, GitLab, and Bitbucket — has transformed code review from a scheduled team activity into a continuous, asynchronous process integrated directly into the development workflow.

Bacchelli and Bird's (2013) large-scale study of code review practices at Microsoft Research remains one of the most comprehensive empirical investigations of the practice. Through developer surveys, interviews, and analysis of code review data, they found that the primary motivations for code review were finding defects, improving code quality, and knowledge transfer — though they also documented that reviewers miss a significant proportion of defects due to time pressure and cognitive limitations. Their findings motivated subsequent research into automated review augmentation.

Tool-assisted static analysis emerged to complement manual review. Tools such as PMD, FindBugs (now SpotBugs), Checkstyle, and SonarQube analyse source code against predefined rule sets, flagging potential violations of coding standards and patterns associated with defects. SonarQube in particular has achieved widespread adoption, supporting over 27 programming languages and providing a web-based dashboard for continuous code quality

monitoring. However, these tools are inherently rule-bound: their rules are defined by domain experts and do not adapt to project-specific patterns. Empirical evaluations consistently report false positive rates above 60%, leading to alert fatigue among developers and, paradoxically, reduced code quality attention as developers learn to ignore automated warnings.

2.3 Machine Learning in Software Engineering

The systematic application of machine learning to software engineering problems — collectively referred to as Mining Software Repositories (MSR) — has grown substantially since the early 2000s. The annual MSR conference, established in 2004, has become a primary venue for research at the intersection of empirical software engineering and data mining.

Nagappan and Ball (2005) provided one of the earliest rigorous demonstrations that software metrics derived from source code structure could predict post-release defect density with statistically significant accuracy. Using relative code churn metrics — measures of how much code changes over time — they showed that high churn is strongly associated with higher defect rates in subsequent releases of Windows Server 2003. Their work established the methodological template for metric-based defect prediction: extract features, train a classifier, evaluate on held-out data.

Zimmermann et al. (2007) extended this line of work by demonstrating that network metrics derived from code dependency graphs — capturing how modules depend on one another — add predictive power beyond traditional complexity metrics. Their analysis of several large Microsoft products showed that highly connected modules are disproportionately likely to contain defects, providing a theoretical foundation for the dependency metrics included in the feature set of this project.

Supervised learning algorithms including Naive Bayes, Logistic Regression, Support Vector Machines, and ensemble methods such as Random Forest have been extensively applied to defect prediction. Lessmann et al. (2008) conducted one of the most comprehensive comparative evaluations, comparing 22 classifiers on 10 public defect datasets from the PROMISE Repository. Their analysis found that while performance differences between classifiers were statistically significant, the practical differences were often modest, and that Random Forest consistently ranked among the top performers. This finding, combined with Random Forest's resistance to overfitting and its built-in feature importance mechanism, motivates its selection in this project.

Transfer learning and cross-project defect prediction have also received significant attention. He et al. (2015) demonstrated that models trained on data from one software project can generalise to others when appropriate instance weighting and transfer strategies are applied, reducing the data collection burden for projects without sufficient historical defect data. More recently, Ni et al. (2022) applied domain adaptation techniques to cross-project prediction, achieving substantial performance improvements over unadapted baselines.

2.4 Bug Prediction Models

Bug prediction has been studied at multiple levels of granularity — file-level, class-level, method-level, and change-level — each with distinct methodological trade-offs. File-level and class-level predictions are the most commonly studied and use metrics from the CK suite (Chidamber and Kemerer object-oriented metrics), Halstead metrics, and McCabe's cyclomatic complexity as primary features.

The CK metric suite, introduced by Chidamber and Kemerer (1994), provides a theoretically grounded set of object-oriented quality attributes: Coupling Between Object classes (CBO), Weighted Methods per Class (WMC), Response For a Class (RFC), Lack of Cohesion in Methods (LCOM), Depth of Inheritance Tree (DIT), and Number of Children (NOC). These metrics have been validated in numerous empirical studies as predictors of fault-proneness, and all six are included in the feature set of the current project.

Process metrics — capturing the development history of a module rather than its static structure — have been shown to be among the most predictive features for defect prediction. Moser et al. (2008) compared process metrics (code churn, number of revisions, number of distinct authors) against product metrics (complexity, size, coupling) and found that process metrics consistently outperformed product metrics on real-world datasets. Incorporating process metrics into the feature set of this project therefore represents an important design decision grounded in empirical evidence.

Change-level prediction, pioneered by Kamei et al. (2013) with the introduction of Just-In-Time (JIT) defect prediction, focuses on individual commits rather than entire files or classes. JIT prediction offers the advantage of providing feedback at the moment code is introduced, enabling immediate correction. However, it requires fine-grained commit-level labelling and features, which increases data collection complexity. The current project operates at the module (file) level, which offers a more tractable data collection pipeline while remaining sufficiently granular for practical utility.

2.5 Deep Learning for Code Analysis

The representational capacity of deep neural networks has motivated their increasingly prominent application to source code analysis. Unlike metric-based approaches that require manual feature engineering, deep learning models can learn representations directly from raw source code tokens, capturing sequential and hierarchical patterns that engineered metrics cannot encode.

White et al. (2016) were among the first to systematically apply Recurrent Neural Networks (RNNs) to code analysis, demonstrating their effectiveness for code clone detection. Their approach represented source code as a sequence of tokens and trained an RNN to learn embeddings that captured semantic similarity between code fragments, outperforming manually engineered feature-based approaches on benchmark datasets.

Long Short-Term Memory (LSTM) networks, introduced by Hochreiter and Schmidhuber (1997), address the vanishing gradient problem that limits the ability of simple RNNs to capture long-range dependencies. LSTMs have proven particularly effective for variable-length sequential data such as tokenised source code. Dam et al. (2018) applied bidirectional LSTMs to vulnerability prediction in C and C++ code, demonstrating that the architecture could learn to identify vulnerability patterns from raw code without hand-crafted rules, achieving competitive performance against both rule-based tools and metric-based ML models.

The advent of transformer-based pre-trained language models has produced a new generation of state-of-the-art code analysis tools. CodeBERT (Feng et al., 2020) is a bimodal pre-trained model for both programming languages and natural language, pre-trained on millions of code-documentation pairs from GitHub. It has achieved top performance on a range of code understanding tasks including defect detection, code search, and clone detection. However, CodeBERT and similar transformer models (GraphCodeBERT, CodeT5) require substantial computational resources for both training and inference, limiting their deployment in resource-constrained environments. This consideration motivates the continued relevance of LSTM-based approaches for production deployment, as employed in this project.

Graph Neural Networks (GNNs) have also been applied to code analysis by exploiting the graph-structured representations inherent to source code: Abstract Syntax Trees (ASTs), Control Flow Graphs (CFGs), and Program Dependence Graphs (PDGs). Allamanis et al. (2018) introduced the Gated Graph Neural Network for variable misuse detection, operating on AST-derived graphs. While these approaches demonstrate strong performance, their adoption is limited by the complexity of graph construction and the computational requirements of GNN training.

2.6 Hybrid and Ensemble Approaches

The observation that metric-based and sequence-based models capture complementary predictive signals has motivated several researchers to explore ensemble and hybrid architectures. Li et al. (2017) combined static code metrics with token-based features in a two-stage prediction pipeline, showing additive performance improvements. Ni et al. (2022) used a deep ensemble combining CNN, LSTM, and metric-based features, demonstrating that model diversity is key to ensemble performance gains.

Despite this evidence, explicit hybrid systems that combine structured software metrics with sequential deep learning in a production-quality deployment pipeline remain relatively rare in the published literature. Most published systems focus on demonstrating model performance in isolation, without addressing the engineering challenges of integrating multiple model types, deploying them through accessible interfaces, and generating actionable, explainable output.

2.7 Explainable AI in Software Engineering

The increasing adoption of machine learning in software engineering has raised important questions about the interpretability and actionability of model predictions. A model that predicts a module as bug-prone without explaining why provides limited practical value to a developer. This has motivated the application of explainability frameworks to defect prediction.

SHAP (SHapley Additive exPlanations), introduced by Lundberg and Lee (2017), provides a unified framework for computing feature attributions based on Shapley values from cooperative game theory. SHAP values are both theoretically well-grounded and practically useful: they assign to each feature a contribution to the individual prediction, enabling both global analysis (which features matter most overall) and local analysis (which features drove a specific prediction). Pornprasit and Tantithamthavorn (2021) demonstrated the utility of SHAP for defect prediction explainability, showing that SHAP-based explanations correlate well with expert developer assessments of fault-prone code characteristics.

2.8 Research Gaps

The preceding review identifies three primary gaps that motivate the design of this project:

- Underexplored hybrid architectures: most published systems address either metric-based prediction or deep learning-based code analysis in isolation. Hybrid approaches that systematically leverage the complementary strengths of both remain underrepresented, particularly in configurations that combine them in a production deployment pipeline.
- Lack of accessible interfaces and actionable output: the majority of published systems report model performance metrics but do not provide accessible APIs or user interfaces. Generating structured, severity-ranked review suggestions — rather than binary defect labels — is rarely addressed.
- Limited explainability integration: while SHAP-based explanations have been studied for defect prediction, few deployed systems integrate explainability into the end-to-end prediction pipeline, from SHAP computation to natural language review suggestion generation.

This project is designed to address all three gaps simultaneously, producing a hybrid, deployable, and explainable system that advances both the research and practical state of automated code quality assessment.

CHAPTER 3: METHODOLOGY

3.1 Overview

This chapter provides a comprehensive description of the methodology employed in the design, development, and evaluation of the AI-Powered Code Review and Bug Prediction System. The methodology is structured around four principal components: (1) a data acquisition and preprocessing pipeline; (2) a feature engineering module; (3) a hybrid machine learning and deep learning prediction engine; and (4) a deployment layer comprising a FastAPI REST backend and an Angular web dashboard. Each component is described in detail, with attention to the design decisions, theoretical motivations, and implementation specifics that characterise each stage.

3.2 System Architecture

The proposed system follows a layered, service-oriented architecture designed to cleanly separate concerns across the data, modelling, and presentation tiers. The four architectural layers are:

- Data Acquisition Layer: responsible for GitHub repository mining, commit labelling using the SZZ algorithm, source code tokenisation for LSTM input, and static metric computation for Random Forest input.
- Feature Engineering Layer: responsible for metric correlation filtering, near-zero variance elimination, z-score normalisation, and SMOTE class balancing on the training split.
- Prediction Engine Layer: comprising the trained Random Forest classifier, the trained LSTM network, the soft voting ensemble integrator, and the SHAP explainability module.
- Deployment Layer: comprising the FastAPI REST backend (exposing /predict/metrics and /predict/code endpoints) and the Angular 16 frontend dashboard (providing code submission, prediction display, and report download functionality).

The prediction engine exposes two primary endpoints. The /predict/metrics endpoint accepts a JSON payload of pre-computed software metrics and returns a bug probability score from the Random Forest model, along with SHAP-ranked feature attributions. The /predict/code endpoint accepts raw source code as a string, performs tokenisation and metric extraction server-side, and returns both LSTM-derived code smell classifications and the combined ensemble prediction with a severity label and natural language review suggestion.

3.3 Data Collection and Preprocessing

3.3.1 Repository Selection

Training and evaluation data were collected by mining open-source GitHub repositories using the GitHub REST API. Repository selection was governed by the following criteria, applied programmatically through API query filters:

7. Minimum commit history: 500 commits (ensuring sufficient historical data for SZZ labelling).
8. Presence of a bug-labelled issue tracker: repositories without a GitHub Issues section or with fewer than 50 issues labelled with bug-related tags were excluded.
9. Minimum codebase size: 1,000 logical lines of code (ensuring sufficient module-level data).
10. Primary language: Python or Java (enabling static metric extraction with Radon and CKJMExtended respectively).
11. Repository activity: last commit within 24 months of data collection (ensuring relevance).

Applying these filters yielded a final dataset of 512 repositories (274 Python, 238 Java) selected from a broader candidate pool of over 4,000 repositories queried using star count and topic filters.

3.3.2 Bug Labelling Using the SZZ Algorithm

Bug-introducing commits were identified using an adaptation of the SZZ algorithm (Śliwerski, Zimmermann, and Zeller, 2005). The procedure was as follows: for each repository, all commits whose messages matched a set of bug-fix indicator regular expressions — including 'fix', 'bug', 'defect', 'issue', 'error', 'patch', 'resolve' — were identified as bug-fix commits. The 'git blame' and 'git log' commands were used to trace back to the commits that introduced the lines modified by each bug-fix commit; these were identified as bug-introducing commits.

For each bug-fix commit, the files modified in that commit were labelled as defect-containing in their pre-fix version. The immediately preceding version of each such file (one commit before the bug-fix) was labelled as 'buggy' (positive class). Files in the same repository that were not associated with any bug-fix commit over the observation period were labelled as 'clean' (negative class). This labelling procedure follows the methodology of He et al. (2015) and Kamei et al. (2013) and is considered a standard approach in the MSR literature.

3.3.3 Source Code Tokenisation for LSTM

Raw source code files were tokenised using language-specific lexers from the Pygments library (version 2.15), which provides tokenisers for over 500 programming languages. For Python files, the Python3Lexer was used; for Java files, the JavaLexer was applied. Tokenisation produced sequences of (token_type, token_value) pairs, from which token values were extracted and mapped to integer indices using a vocabulary of 10,000 tokens constructed from the training corpus using frequency-based selection. Out-of-vocabulary tokens were mapped to a special UNK (unknown) token with index 0.

Sequences were padded with a special PAD token (index 1) or truncated to a fixed maximum length of 512 tokens for LSTM input. The choice of 512 tokens was informed by an analysis of the distribution of file lengths in the training corpus: the 75th percentile of file lengths was 487 tokens, and 512 was selected as a round number slightly above this threshold to capture the majority of file content without excessive memory consumption.

3.4 Feature Engineering

3.4.1 Software Metric Extraction

For the Random Forest model, 87 software metrics were computed per code module using two static analysis tools: Radon (version 5.1) for Python modules and CKJMExtended (version 2.3) for Java modules. These metrics span five categories as follows:

Category	Example Metrics	Tool
Size Metrics	LOC, LLOC, number of methods, attributes, blank lines	Radon / CKJM
Complexity Metrics	Cyclomatic complexity, Halstead volume, effort, difficulty, bugs estimate	Radon
OO Metrics (CK Suite)	CBO, WMC, RFC, LCOM, DIT, NOC	CKJMExtended
Process Metrics	Code churn, number of commits, distinct authors, prior defect count, module age	Git log
Dependency Metrics	Fan-in, fan-out, instability index (fan-out / (fan-in + fan-out))	Custom parser

Table 3.2 – Software metric categories: feature groups, tools, and example metrics

3.4.2 Feature Selection

Of the 87 initially computed metrics, 26 were removed through two selection steps. First, all pairs of features with Pearson correlation coefficient $|r| > 0.9$ were identified (18 pairs found); from each pair, the feature with lower individual predictive power (assessed by univariate AUC-ROC on the training set) was removed, yielding a reduced set. Second, features with near-zero variance (defined as a ratio of the frequency of the most common value to the frequency of the second most common value greater than 95/5, following the caret convention) were removed, eliminating a further 8 features. The final feature set contained 61 features used for Random Forest model training.

All 61 features were standardised using z-score normalisation (subtracting the training set mean and dividing by the training set standard deviation). Normalisation parameters were computed exclusively on the training set and applied to validation and test sets to prevent data leakage.

3.4.3 Class Imbalance Handling

The raw labelled dataset exhibited significant class imbalance: 25.8% buggy modules and 74.2% clean modules. Training a classifier on such an imbalanced dataset without correction risks producing a model that is biased toward predicting the majority (clean) class. To address this, Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training set.

SMOTE generates synthetic minority class (buggy) instances by interpolating between existing minority class instances and their k-nearest neighbours (k=5 used). The class ratio in the training set was balanced to approximately 50:50 after SMOTE application. Critically, SMOTE was applied only after the data was split into training, validation, and test sets, ensuring that the validation and test sets reflect the original class distribution (25.8% buggy) to provide realistic evaluation metrics.

3.5 Model Development

3.5.1 Random Forest Classifier

A Random Forest classifier was implemented and trained using Scikit-learn (version 1.3). Random Forest is an ensemble of decision trees where each tree is trained on a bootstrap sample of the training data and evaluates a random subset of features at each split (the 'random subspace' method). The final prediction is produced by averaging the predicted class probabilities across all trees, providing both a bug probability score and a binary classification.

Hyperparameter optimisation was performed via 5-fold stratified cross-validation grid search over the following parameter space:

- Number of estimators (trees): {100, 200, 500}
- Maximum tree depth: {None (unlimited), 10, 20, 30}
- Minimum samples required to split a node: {2, 5, 10}
- Class weight: {'balanced' (inverse class frequency weighting), None}

The grid search evaluated 3 × 4 × 3 × 2 = 72 hyperparameter combinations, with each combination evaluated using mean AUC-ROC across the 5 folds as the optimisation criterion. The optimal configuration identified was: 200 estimators, no maximum depth constraint (unlimited), minimum samples split of 5, and balanced class weighting. Training time for the optimal configuration on the full training set was approximately 4.2 minutes on the evaluation hardware.

Feature importance was computed using the Random Forest's built-in mean decrease in impurity (MDI) measure, and SHAP TreeExplainer was applied post-training to compute SHAP values for individual predictions.

3.5.2 LSTM Network Architecture

The LSTM model was implemented using TensorFlow 2.12 / Keras. The architecture was designed to balance representational capacity against computational tractability for deployment, and comprises the following layers:

12. Embedding Layer: Maps integer token indices to dense 128-dimensional embedding vectors. Vocabulary size: 10,000 tokens. The embedding matrix is learned jointly with the rest of the network during training. Total parameters: 1,280,000.

13. First LSTM Layer: 128 LSTM units with return_sequences=True (passing the full sequence of hidden states to the next layer). Dropout (p=0.3) applied between time steps. Total parameters: 132,608.
14. Second LSTM Layer: 64 LSTM units with return_sequences=False (returning only the final hidden state). Dropout (p=0.3) applied between time steps. Total parameters: 49,408.
15. Dense Hidden Layer: 64 units with ReLU activation. Dropout (p=0.2) applied before the output layer. Total parameters: 4,160.
16. Output Layer: 1 unit with sigmoid activation, producing a scalar bug probability in [0,1]. Total parameters: 65.

Total trainable parameters: 1,466,241. The model was trained using the Adam optimiser (learning rate: 0.001, β1=0.9, β2=0.999) with binary cross-entropy loss. Training ran for up to 30 epochs with early stopping (patience: 5 epochs, monitoring validation loss, restoring best weights). The batch size was set to 64. Training completed in approximately 2.8 hours on the NVIDIA GeForce GTX 1650 GPU.

3.5.3 Ensemble Integration

The final prediction for each module is produced by a soft voting ensemble combining the probability outputs of the Random Forest and LSTM models. Soft voting averages the predicted class probabilities (rather than the hard class labels) of the constituent models, which is generally superior to hard voting because it exploits the calibrated probability estimates of each model.

The ensemble prediction is computed as:

$$P(buggy) = w_RF \times P_RF(buggy) + w_LSTM \times P_LSTM(buggy)$$

where w_RF = 0.55 and w_LSTM = 0.45. These weights were determined empirically by evaluating all combinations of weights in increments of 0.05 on the validation set, selecting the combination that maximised AUC-ROC. A classification threshold of 0.5 is applied to P(buggy) to produce the final binary prediction.

The weighting slightly favours the Random Forest model (0.55 vs 0.45), consistent with its marginally higher individual performance on the validation set (AUC-ROC: 0.91 vs 0.90). However, the performance improvement of the ensemble over either individual model is primarily attributable to the diversity of their representations — the Random Forest exploits structured metrics while the LSTM exploits sequential code patterns — rather than to the precise weight ratio.

3.6 Explainability with SHAP

Model explainability is implemented using SHAP (SHapley Additive exPlanations), which provides a theoretically principled framework for assigning feature contributions to individual predictions. SHAP values are computed as the average marginal contribution of each feature across all possible feature orderings, drawing on Shapley values from cooperative game theory.

For the Random Forest model, SHAP TreeExplainer is used, which exploits the tree structure to compute exact SHAP values in polynomial time. For each prediction, the top features by absolute SHAP value are identified and ranked. The natural language review suggestion is generated by mapping the top SHAP features to a template-based description:

- If CBO is in the top-3 features and its SHAP value is positive: the suggestion includes a note on high inter-module coupling and recommends refactoring toward single-responsibility classes.
- If WMC is in the top-3 features: the suggestion notes excessive method complexity and recommends decomposition.
- If code churn is highly predictive: the suggestion notes high recent modification activity as a risk indicator.

This template-based approach produces deterministic, reproducible suggestions that are grounded in the specific predictive signals driving each individual prediction, rather than generic advice.

3.7 Code Smell Detection

Code smell detection is implemented as a multi-label classification task performed by the LSTM model. The model is trained to simultaneously detect five categories of code smells:

17. God Class: a class that has grown to encompass too many responsibilities, identified by high WMC, high LOC, and high CBO.
18. Long Method: a method exceeding a defined complexity and length threshold, identified by high cyclomatic complexity and high method-level LOC.
19. Duplicate Code: structurally similar code appearing in multiple locations, identified by token-level similarity analysis during preprocessing.
20. Feature Envy: a method that accesses data from another class more than its own, identified by high RFC and high inter-class coupling.
21. Data Class: a class that contains only fields and accessors without meaningful behaviour, identified by low WMC and high number of attributes.

Code smell labels were generated from the training corpus using established threshold-based heuristics (Palomba et al., 2014) applied to the computed metrics, enabling the LSTM to learn the relationship between raw code token patterns and these higher-level structural quality violations.

3.8 System Implementation

3.8.1 Backend API (FastAPI)

The backend REST API was implemented using FastAPI (version 0.103), a modern Python web framework that provides automatic request validation, OpenAPI schema generation, and asynchronous request handling. The API is served using Uvicorn (version 0.23) as the ASGI server.

At startup, model artefacts are loaded into memory: the Random Forest model is deserialised from a joblib file (serialised via joblib 1.3), and the LSTM model is loaded from a TensorFlow SavedModel directory. Maintaining models in memory eliminates per-request model loading overhead, reducing inference latency to tens of milliseconds for metric-based requests and under 200ms for code-based requests on the evaluation hardware.

The API exposes the following primary endpoints:

- /predict/metrics (POST): Accepts a JSON payload containing a flat dictionary of 61 software metric values. Returns: bug_probability (float), bug_label (bool), shap_values (dict mapping feature names to SHAP values), review_suggestion (string), severity (string: Low / Medium / High / Critical).
- /predict/code (POST): Accepts a JSON payload containing source_code (string) and language ('python' or 'java'). Server-side metric extraction and tokenisation are performed. Returns: ensemble_bug_probability, bug_label, code_smells (list), severity, review_suggestion, and contributing_metrics.
- /health (GET): Returns API health status and model version information.

3.8.2 Frontend Dashboard (Angular 16)

The Angular 16 frontend provides a web-based interface for interacting with the prediction API. The dashboard is structured around three primary views:

22. Code Submission View: A Monaco Editor-based code input panel (providing syntax highlighting for Python and Java), a language selector, and a Submit for Analysis button that sends a POST request to the /predict/code endpoint.
23. Results View: Displays the prediction result with a colour-coded severity badge (green: Low, yellow: Medium, orange: High, red: Critical), the ensemble bug probability as a gauge chart, detected code smells as labelled tags, and a SHAP-ranked feature contribution bar chart visualising the top contributing metrics.
24. Report View: Renders the structured review report with the natural language suggestion, metric values, and code smell descriptions. Provides download buttons for both JSON and PDF report formats (PDF generation uses jsPDF on the frontend).

3.9 Dataset Statistics

The final dataset comprised 18,742 labelled code modules derived from 512 GitHub repositories. Table 3.5 summarises the dataset statistics across the three experimental splits.

Split	Total Modules	Buggy	Clean	Buggy %
Training (70%)	13,119	3,384	9,735	25.8%
Validation (15%)	2,812	725	2,087	25.8%
Test (15%)	2,811	725	2,086	25.8%
Total (pre-SMOTE)	18,742	4,834	13,908	25.8%

Training (post-SMOTE)	**19,470**	9,735	9,735	50.0%

Table 3.5 – Dataset statistics: class distribution across splits before and after SMOTE

CHAPTER 4: RESULTS AND EVALUATION

4.1 Experimental Setup

All experiments were conducted on a workstation equipped with an Intel Core i7-10750H CPU (6 cores, 12 threads, base clock 2.6 GHz, boost clock 5.0 GHz), 16 GB DDR4 RAM, and an NVIDIA GeForce GTX 1650 GPU (4 GB GDDR6 VRAM), running Ubuntu 22.04 LTS. The LSTM model was trained using TensorFlow's GPU acceleration via CUDA 11.8 and cuDNN 8.6. All metric computation, preprocessing, and Random Forest training were performed on the CPU. Python 3.10 was used throughout; key dependencies and versions are listed in Appendix A.

Evaluation metrics reported are: accuracy (proportion of correct predictions), precision (positive predictive value), recall (sensitivity / true positive rate), F1-score (harmonic mean of precision and recall), and AUC-ROC (area under the receiver operating characteristic curve). All metrics are computed on the held-out test set (n = 2,812 modules) using the original, unbalanced class distribution (25.8% buggy). The threshold for binary classification is set at 0.5 for all models.

4.2 Model Performance

Table 4.1 summarises the performance of each model component and the final ensemble on the held-out test set. All values are reported to two decimal places.

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Random Forest	83.4%	0.81	0.86	0.83	0.91
LSTM (Sequential)	81.7%	0.79	0.88	0.83	0.90
Logistic Regression (baseline)	76.3%	0.74	0.75	0.74	0.82
Naive Bayes (baseline)	74.1%	0.71	0.72	0.71	0.80
Proposed Ensemble (Ours)	**87.1%**	0.84	0.89	0.87	**0.94**

Table 4.1 – Model performance on the held-out test set (n = 2,812 modules)

The proposed ensemble achieves the highest performance across all five metrics. The 3.7 percentage point accuracy improvement over the stronger individual model (Random Forest at 83.4%) and the 5.4 point AUC-ROC improvement over Random Forest (0.94 vs 0.91) confirm that the two model types capture complementary predictive signals that are additive when combined. The LSTM's higher recall (0.88) relative to its precision (0.79) and the Random Forest's more balanced profile (0.81 precision, 0.86 recall) suggest that the models have

complementary error characteristics: the LSTM is more sensitive (fewer false negatives) while the Random Forest is more specific (fewer false positives). The ensemble benefits from both properties.

The higher recall (0.89) relative to precision (0.84) in the final ensemble reflects a deliberate design decision. In a code review context, a false negative (predicting a buggy module as clean, thereby missing a real defect) has a higher cost than a false positive (flagging a clean module for unnecessary review). The ensemble is therefore intentionally tuned to favour recall over precision, consistent with the application domain requirements.

4.3 Comparison with Baseline Approaches

Table 4.2 compares the proposed system against rule-based static analysis tools and classical machine learning baselines, all evaluated on the same test set.

System	Accuracy	F1-Score	AUC-ROC
SonarQube (rule-based)	71.2%	0.68	N/A
PMD (rule-based)	68.5%	0.65	N/A
FindBugs / SpotBugs (rule-based)	69.8%	0.66	N/A
Logistic Regression (61 metrics)	76.3%	0.74	0.82
Naive Bayes (61 metrics)	74.1%	0.71	0.80
SVM (61 metrics)	78.4%	0.76	0.84
Random Forest (61 metrics)	83.4%	0.83	0.91
LSTM only (tokenised code)	81.7%	0.83	0.90
Proposed Ensemble (Ours)	**87.1%**	0.87	0.94

Table 4.2 – Comparison with baseline approaches on the same test set

The proposed ensemble outperforms all evaluated baselines across all reported metrics. The improvement over rule-based static analysis tools is particularly substantial: 15.9 percentage points over SonarQube and 18.6 points over PMD in accuracy. This confirms that data-driven prediction captures defect-indicative patterns that static rule sets cannot encode.

Among ML baselines, SVM provides the strongest performance at 78.4% accuracy and AUC-ROC 0.84, followed by Random Forest at 83.4% and AUC-ROC 0.91. The ensemble's improvement of 3.7 points in accuracy and 0.03 in AUC-ROC over Random Forest alone demonstrates that the LSTM's sequential code representations provide incremental value even above a strong structured-feature baseline.

4.4 SHAP Feature Importance Analysis

SHAP TreeExplainer was applied to the Random Forest model to compute feature attribution values across the entire test set. Table 4.3 lists the top-10 features by mean absolute SHAP value, representing the features with the greatest average impact on the model's predictions.

Rank	Feature	Mean |SHAP|	Category
1	**CBO (Coupling Between Object Classes)**	0.341	OO Metrics
2	**WMC (Weighted Methods per Class)**	0.283	OO Metrics
3	**LOC (Lines of Code)**	0.191	Size Metrics
4	**Code Churn (last 6 months)**	0.187	Process Metrics
5	**Cyclomatic Complexity (avg. per method)**	0.176	Complexity Metrics
6	**RFC (Response For a Class)**	0.168	OO Metrics
7	**Fan-Out**	0.152	Dependency Metrics
8	**Number of Prior Defects**	0.143	Process Metrics
9	**Halstead Difficulty**	0.138	Complexity Metrics
10	**Number of Distinct Authors**	0.122	Process Metrics

Table 4.3 – Top-10 predictive features by mean absolute SHAP value

The dominance of CBO and WMC in the SHAP ranking is consistent with prior empirical literature. High inter-module coupling (CBO) increases the probability that a defect in one module propagates to dependent modules, and high method complexity (WMC) is a well-established predictor of fault-proneness. The strong showing of process metrics (Code Churn, Number of Prior Defects, Number of Distinct Authors) corroborates Moser et al.'s (2008) finding that change history is among the most predictive feature categories.

4.5 Code Smell Detection Results

Table 4.4 presents the LSTM model's performance on the multi-label code smell detection task, evaluated on the held-out test set.

Code Smell	Prevalence	Precision	Recall	F1	AUC
God Class	12.4%	0.82	0.79	0.80	0.91
Long Method	18.7%	0.78	0.83	0.80	0.89
Duplicate Code	9.2%	0.71	0.76	0.73	0.85

Feature Envy	7.8%	0.74	0.71	0.72	0.83
Data Class	6.1%	0.76	0.68	0.72	0.84

Table 4.4 – Code smell detection performance by category (LSTM model, test set)

4.6 System Output Example

The following illustrates a representative system output for a flagged Python module (a utility class from one of the mined repositories, anonymised):

- Module: utils/data_processor.py (anonymised repository)
- Ensemble bug probability: 0.91 | Severity: CRITICAL
- Detected code smells: God Class, Long Method, Feature Envy
- Top contributing features (SHAP): CBO = 18 (ΔP = +0.34), WMC = 24 (ΔP = +0.28), LOC = 847 (ΔP = +0.19), Code Churn = 1,247 lines in 6 months (ΔP = +0.15)
- Generated review suggestion: "This module exhibits critically high inter-module coupling (CBO = 18, threshold: 10) and excessive method complexity (average WMC = 24, threshold: 15). The module has been substantially modified in the past 6 months (1,247 lines churned), which is a strong process-level risk indicator. Recommended actions: (1) Decompose into smaller, single-responsibility classes; (2) Reduce inter-module dependencies through dependency inversion; (3) Extract frequently modified methods into a dedicated utility class. Priority: CRITICAL."

4.7 Inference Latency

Table 4.5 presents average inference latency benchmarks measured over 100 consecutive requests on the evaluation hardware, reporting mean and 95th percentile response times.

Request Type	Mean Latency (ms)	P95 Latency (ms)
/predict/metrics (RF only)	38	61
/predict/code (LSTM + RF ensemble)	167	214
Metric extraction (server-side)	89	127
SHAP computation (per request)	23	38

Table 4.5 – Inference latency benchmarks

The /predict/metrics endpoint achieves a mean latency of 38ms, well within the sub-100ms threshold typically associated with interactive responsiveness. The /predict/code endpoint, which performs server-side metric extraction and LSTM inference, achieves a mean latency of 167ms, acceptable for a non-interactive batch review context. Both endpoints are suitable for integration into continuous integration pipelines.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Conclusion

This project has presented the design, implementation, and empirical evaluation of an AI-Powered Code Review and Bug Prediction System — a comprehensive, data-driven solution that addresses the documented limitations of existing rule-based static analysis tools through a novel hybrid machine learning and deep learning architecture.

The central contribution of this work is a soft voting ensemble that combines a Random Forest classifier trained on 61 engineered software metrics with an LSTM network operating directly on tokenised source code sequences. The ensemble achieves 87.1% accuracy and an AUC-ROC of 0.94 on a held-out test set of 2,812 labelled code modules — outperforming all evaluated baselines, including SonarQube (+15.9% accuracy), PMD (+18.6%), Logistic Regression (+10.8%), Naive Bayes (+13.0%), and SVM (+8.7%). The performance improvement over either individual constituent model confirms that the two modalities capture complementary predictive signals: the Random Forest exploits the rich structural information encoded in software metrics, while the LSTM captures sequential code patterns that metrics cannot represent.

Beyond raw prediction performance, this project addresses the practical limitations that constrain the utility of published research systems. SHAP-based explainability translates model predictions into metric-level feature attributions and natural language review suggestions ranked by severity, providing actionable guidance that a developer can act upon directly. The FastAPI REST backend and Angular 16 dashboard make the system accessible without requiring specialist ML knowledge, enabling integration into development workflows through standard HTTP requests.

The dataset of 18,742 labelled code modules mined from 512 GitHub repositories provides a reproducible benchmark for future comparative evaluations, and the full experimental pipeline is documented in sufficient detail for replication.

5.2 Limitations

Several limitations of the current implementation should be acknowledged:

- Language scope: the system was trained and evaluated exclusively on Python and Java codebases. Generalisation to other languages (C++, JavaScript, Go, Rust) requires language-specific tokenisers and static analysis tools, as well as retraining or fine-tuning on language-specific data.
- Fixed-length LSTM input: the LSTM model requires sequences padded or truncated to 512 tokens. For very large files (above the 75th percentile of file lengths), truncation may result in

loss of information from later sections of the file, potentially affecting prediction accuracy for large modules.

- Module-level analysis: the system analyses entire files (modules) rather than individual commits or code diffs. This limits its utility in pull request review workflows, where change-level feedback on specific newly added lines is most actionable.
- LSTM explainability: SHAP TreeExplainer is applicable to the Random Forest component only. Unified explainability across both model types — providing token-level attribution from the LSTM alongside metric-level attribution from the Random Forest — is not yet implemented.
- Single-label ensemble output: the ensemble produces a single bug probability from both models. A more sophisticated integration might separately weight the models' contributions for specific metric ranges or code contexts.
- Evaluation on public repositories only: the dataset consists entirely of open-source GitHub repositories. Performance on proprietary industrial codebases, which may have different coding standards, team practices, and defect characteristics, has not been evaluated.

5.3 Future Work

The following directions are identified for future development and research:

25. Transformer-based code models: Integrate CodeBERT or GraphCodeBERT as an additional ensemble component. These models leverage pre-training on massive code corpora and have demonstrated state-of-the-art performance on a range of code understanding tasks. Combining a transformer encoder with the existing Random Forest and LSTM components could yield further performance gains.
26. Change-level (Just-In-Time) prediction: Extend the system to operate on commit diffs rather than full module snapshots, enabling prediction at the granularity of individual code changes. This would allow integration directly into pull request review workflows and provide more timely feedback to developers.
27. Multi-language support: Extend the pipeline to support C++, JavaScript, TypeScript, Go, and Rust by integrating language-specific tokenisers (e.g., Clang for C++, Esprima for JavaScript) and appropriate static analysis tools.
28. IDE plugin development: Develop plugins for VS Code and JetBrains IDEs (IntelliJ IDEA, PyCharm) that surface predictions directly in the developer's editing environment, providing inline annotations without requiring interaction with the web dashboard.
29. Unified explainability framework: Investigate Integrated Gradients or LIME (Local Interpretable Model-agnostic Explanations) as explainability methods applicable to the LSTM component, enabling consistent feature attribution across both model types and potentially providing token-level explanations.
30. Graph-based code representations: Explore GNN-based approaches operating on AST, CFG, or PDG representations of code, which capture structural relationships not available in the token sequence or flat metric representations used by the current system.
31. Controlled user study: Conduct a randomised controlled study with professional software developers to evaluate the practical utility, usability, and trust impact of the system's generated review suggestions compared to baseline tools.

32. Continuous learning pipeline: Implement an active learning component that incorporates developer feedback (accepting or rejecting predictions) to continuously update the model, adapting to project-specific coding patterns over time.

CHAPTER 3 (CONTINUED): DETAILED SYSTEM DESIGN

3.10 Detailed Data Pipeline Implementation

3.10.1 GitHub API Integration

The data collection pipeline was implemented as an automated Python script using the PyGitHub library (version 1.59.1), which provides a Pythonic wrapper around the GitHub REST API v3. Repository discovery was performed in two phases. In the first phase, repositories were enumerated using the /search/repositories endpoint with language and topic filters. Queries were batched in groups of 30 (the maximum permitted by the API's pagination) and rate limiting was handled through exponential backoff: on receipt of a 403 or 429 rate-limit response, the pipeline paused for the duration specified in the X-RateLimit-Reset header plus a five-second safety margin.

In the second phase, each candidate repository was individually assessed against the selection criteria (minimum 500 commits, minimum 50 bug-labelled issues, minimum 1,000 LLOC). Repositories failing any criterion were discarded. The assessment phase consumed approximately 14 hours of wall-clock time for the 4,000-repository candidate pool, due to per-repository API calls required for commit count and issue label enumeration.

For each selected repository, the full commit history was cloned locally using GitPython. Commit metadata (author, timestamp, message, modified files) was extracted from the local clone, avoiding further API calls and operating at local disk I/O speed. The clone-and-process pipeline was parallelised across 8 worker processes using Python's multiprocessing module, reducing total data collection time to approximately 31 hours for the 512-repository corpus.

3.10.2 SZZ Algorithm Implementation Details

The SZZ algorithm was implemented as follows. For each repository, commit messages were scanned using the following compiled regular expression:

```
r'\b(fix|bug|defect|issue|error|patch|resolve|repair|correct|wrong|fail|cr
ash|broken)\b'
```

Commits whose messages matched this pattern were classified as bug-fix commits. For each bug-fix commit, the set of modified files was obtained from the commit diff. For each such file, 'git blame' was applied to the pre-fix version of the file, annotating each line with the commit that last modified it. Lines that were deleted or modified by the bug-fix commit (as identified by the unified diff) were traced back to their introducing commits using git blame. These introducing commits were classified as bug-introducing commits, and the files they modified were labelled buggy.

One known limitation of SZZ is the introduction of noise through meta-commits (e.g., merge commits, formatting-only commits). To mitigate this, commits that modified more than 50 files were excluded as likely refactoring or merge commits, following the heuristic of Kamei et al. (2013). Additionally, commits modifying only non-source files (documentation, configuration, build scripts) were excluded from labelling.

3.10.3 Static Metric Computation Pipeline

Python metric computation was performed using Radon's programmatic API. For each Python source file, the following metrics were computed: raw metrics (LOC, LLOC, SLOC, comments, multi-line strings, blank lines) via radon.raw; cyclomatic complexity per function and class method via radon.complexity; Halstead metrics (volume, difficulty, effort, bugs, time) via radon.metrics; and maintainability index via radon.metrics. File-level aggregates (sum, mean, max) were computed for method-level metrics.

Java metric computation used CKJMExtended, a command-line tool that computes the full CK metric suite (CBO, WMC, RFC, LCOM, DIT, NOC) plus extensions (Ca, Ce, NPM, LCOM3, CAM, IC, CBM, AMC) for each Java class. CKJMExtended was invoked via subprocess for each compiled Java class file, with compilation performed using javac. Compilation failures (due to missing dependencies in the cloned repositories) were handled by skipping the affected files and logging the failure; approximately 8.3% of Java files failed to compile and were excluded from the dataset.

3.11 Model Training Details

3.11.1 LSTM Training Curves and Convergence

The LSTM model was trained for 30 epochs with early stopping applied with a patience of 5 epochs monitoring validation loss. Training converged at epoch 22 (early stopping triggered at epoch 27, restoring weights from epoch 22). The final training loss was 0.2841 and validation loss was 0.3107, indicating moderate generalisation without severe overfitting. The validation AUC-ROC stabilised at 0.91 by epoch 18 and showed no significant improvement thereafter.

The learning rate was kept constant at 0.001 throughout training. Experiments with learning rate scheduling (cosine annealing, step decay) did not yield significant performance improvements on the validation set and were not included in the final configuration. Gradient clipping (max norm: 1.0) was applied to stabilise training, as preliminary experiments without clipping exhibited occasional loss spikes in early epochs.

3.11.2 Hyperparameter Sensitivity Analysis

An ablation study was conducted to assess the sensitivity of the LSTM model to its principal hyperparameters. The following variations were evaluated on the validation set, with all other parameters held at their final configuration:

Hyperparameter	Values Tested	Best Value	Val AUC-ROC
Embedding dimension	64, 128, 256	128	0.903
LSTM units (layer 1)	64, 128, 256	128	0.903
Dropout rate	0.1, 0.2, 0.3, 0.4	0.3	0.903
Sequence length	256, 512, 768	512	0.903
Batch size	32, 64, 128	64	0.903
Learning rate	0.0001, 0.001, 0.01	0.001	0.903

Table 3.6 – LSTM hyperparameter sensitivity analysis on validation set

The results indicate that the model is relatively robust to hyperparameter variation within the tested ranges, with AUC-ROC varying by no more than 0.012 across all configurations. The selected configuration (embedding dim 128, 128 LSTM units, dropout 0.3, sequence length 512, batch size 64, learning rate 0.001) represents the optimal point across the combined search.

3.12 API Design and Endpoint Specification

The FastAPI backend exposes a fully documented OpenAPI (Swagger) schema, auto-generated from type-annotated endpoint function signatures. The complete endpoint specification is as follows:

3.12.1 POST /predict/metrics

Request body (JSON): A flat dictionary of 61 software metric values, with string keys corresponding to metric names and float values. Example partial payload:

```
{"cbo": 12, "wmc": 18, "rfc": 34, "lcom": 0.72, "loc": 312,
"cyclomatic_complexity": 8.4, ...}
```

Response body (JSON): bug_probability (float, 0–1), bug_label (bool), severity (string: Low/Medium/High/Critical), shap_values (object: feature name → SHAP value), top_features (array of top-5 feature objects with name, value, shap), review_suggestion (string), processing_time_ms (float).

3.12.2 POST /predict/code

Request body (JSON): source_code (string, required), language (string: ‘python’ or ‘java’, required), filename (string, optional, used for display in the report).

Response body (JSON): ensemble_bug_probability, bug_label, severity, code_smells (array of objects: name, confidence, description, recommendation), contributing_metrics (object), review_suggestion (string), processing_time_ms.

Error responses: 400 Bad Request (unsupported language, malformed JSON), 422 Unprocessable Entity (missing required fields), 500 Internal Server Error (metric extraction or inference failure, with error details in development mode).

3.12.3 Rate Limiting and Security

The production deployment includes rate limiting at 60 requests per minute per IP address, enforced using the slowapi library. CORS (Cross-Origin Resource Sharing) is configured to allow requests from the Angular frontend domain only. Input validation is enforced through Pydantic models: the metrics endpoint validates that all 61 required metric keys are present and that all values are finite floats; the code endpoint enforces a maximum source code length of 50,000 characters.

3.13 Frontend Architecture Details

3.13.1 Component Structure

The Angular 16 frontend is structured around six principal components organised in a feature module architecture:

33. AppComponent: Root component providing the navigation shell, route outlet, and global state management via a shared AnalysisService.
34. CodeEditorComponent: Wraps the Monaco Editor (version 0.44) for syntax-highlighted code input. Supports Python and Java syntax highlighting, configurable editor theme (light/dark), and keyboard shortcuts for analysis submission (Ctrl+Enter).
35. AnalysisResultComponent: Displays the prediction result with a severity badge (colour-coded via CSS custom properties), ensemble probability displayed as an ngx-gauge arc widget, and a Chart.js horizontal bar chart for SHAP feature contributions.
36. CodeSmellsComponent: Renders detected code smells as a grid of card components, each displaying the smell name, confidence score, and a collapsible description and recommendation panel.
37. MetricsDashboardComponent: Provides an alternative input mode accepting individual metric values through a form, calling the /predict/metrics endpoint directly.
38. ReportDownloadComponent: Generates and downloads review reports. JSON download uses the browser's Blob API. PDF generation uses jsPDF (version 2.5) with autoTable plugin for tabular metric data.

3.13.2 State Management and HTTP Layer

Application state is managed through RxJS Observables using Angular's reactive forms and the HttpClient module. The AnalysisService exposes two public methods — analyseCode(code, language) and analyseMetrics(metrics) — each returning an Observable that wraps the corresponding HTTP POST request. Loading state, error handling, and result caching are managed via BehaviorSubjects, enabling reactive UI updates without manual change detection cycles.

HTTP interceptors handle two cross-cutting concerns: (1) the AuthInterceptor attaches a static API key header (configurable via environment.ts) to all requests; (2) the ErrorInterceptor catches HTTP errors and maps them to user-friendly error messages displayed in a toast notification component.

CHAPTER 6: TESTING AND VALIDATION

6.1 Overview

This chapter describes the testing strategy employed throughout the development of the system. Testing was conducted at three levels: unit testing of individual components, integration testing of the API and data pipeline, and system-level validation through the empirical evaluation described in Chapter 4. The goal of the testing strategy was to ensure correctness of the implementation, reproducibility of the empirical results, and reliability of the deployed system under realistic operating conditions.

6.2 Unit Testing

6.2.1 Data Pipeline Tests

Unit tests for the data collection and preprocessing pipeline were implemented using Python's pytest framework (version 7.4). The test suite comprises 47 test cases organised into four test modules:

- test_szz.py (12 tests): Validates the SZZ labelling implementation against a synthetic repository with manually annotated bug-fix and bug-introducing commits. Verified that: (a) commits matching the bug-fix regex are correctly identified; (b) git blame traces correctly to the introducing commit; (c) meta-commit exclusion filters are applied correctly; (d) edge cases (empty commits, binary files, renamed files) are handled gracefully.
- test_tokeniser.py (9 tests): Validates source code tokenisation for Python and Java inputs, including: correct vocabulary mapping, OOV handling, sequence padding and truncation, and handling of empty and malformed inputs.
- test_metrics.py (18 tests): Validates metric extraction for both Python (Radon) and Java (CKJMExtended) inputs against known reference values computed manually for a set of synthetic code modules. Verified metric values within a tolerance of 1% of reference.
- test_features.py (8 tests): Validates feature selection, correlation filtering, normalisation, and SMOTE application, including: correct feature removal when correlation exceeds threshold; normalisation parameter persistence between train and test sets; and SMOTE application only to the training set.

All 47 unit tests pass on the final codebase. Test execution time is under 3 minutes on the evaluation hardware.

6.2.2 Model Tests

Model-level tests (11 tests in test_models.py) validated:

- Input/output shape correctness for the LSTM model across batch sizes of 1, 16, and 64.
- Random Forest prediction outputs are valid probabilities in [0,1] for all test inputs.
- Ensemble weights sum to 1.0 and produce valid probability outputs.

- SHAP values are finite and their sum approximates the difference between the prediction and the model's base value (within a tolerance of 0.01).
- Severity thresholds (Low: <0.3, Medium: 0.3–0.5, High: 0.5–0.75, Critical: >0.75) are correctly applied.

6.3 Integration Testing

6.3.1 API Integration Tests

Integration tests for the FastAPI backend were implemented using pytest combined with FastAPI's built-in TestClient, which runs the application in-process without requiring a network socket. The integration test suite (test_api.py, 23 tests) validates:

39. The /predict/metrics endpoint returns a valid response with all expected fields for a well-formed request with all 61 metrics present.
40. The /predict/code endpoint returns valid responses for Python and Java inputs of varying lengths (10 lines, 100 lines, 500 lines).
41. Both endpoints return HTTP 400 for malformed JSON inputs.
42. Both endpoints return HTTP 422 for inputs missing required fields.
43. The /health endpoint returns HTTP 200 with model version information.
44. Rate limiting returns HTTP 429 after the threshold is exceeded.
45. CORS headers are correctly set for requests from the allowed frontend origin.
46. Inference is deterministic: repeated requests with identical inputs return identical outputs.

6.3.2 End-to-End Pipeline Tests

End-to-end tests validated the complete pipeline from raw source code input to structured review report output for four representative test cases:

- A known-buggy Python class from the test corpus (labelled buggy by SZZ): verified that the system predicts bug_label=true with probability above 0.7.
- A known-clean Python module from the test corpus: verified that the system predicts bug_label=false with probability below 0.3.
- A synthetically constructed 'God Class' (high WMC, high CBO, high LOC): verified that the God Class code smell is detected and that the review suggestion references coupling and complexity metrics.
- A minimal Java class with no methods (Data Class pattern): verified that the Data Class smell is detected.

6.4 Performance and Load Testing

Load testing was conducted using Locust (version 2.17), an open-source load testing framework. The load test simulated concurrent users submitting code analysis requests at a rate of up to 50 requests per second for a duration of 5 minutes. Results:

Concurrency	RPS	P50 (ms)	P95 (ms)	Error Rate

1 user	1	42	67	0%
5 concurrent users	5	48	89	0%
10 concurrent users	10	63	142	0%
25 concurrent users	22	112	287	0%
50 concurrent users	31	198	481	2.1%

Table 6.1 – Load test results: API throughput and latency under concurrent load

The system maintains sub-200ms P50 latency up to 25 concurrent users. At 50 concurrent users, the 2.1% error rate is attributable to rate limiting (HTTP 429 responses) rather than application errors, confirming the rate limiter is functioning as intended. For the expected usage pattern (individual developers or small teams), the system comfortably handles concurrent load.

6.5 Cross-Validation Results

In addition to the train/validation/test split evaluation reported in Chapter 4, a 10-fold stratified cross-validation was conducted on the full dataset (excluding the held-out test set) to provide a more robust estimate of generalisation performance. Results:

Model	CV Accuracy	CV Precision	CV Recall	CV F1	CV AUC-ROC
Random Forest	82.8 ± 1.2%	0.80 ± 0.02	0.85 ± 0.03	0.82 ± 0.02	0.90 ± 0.01
LSTM	81.1 ± 1.8%	0.78 ± 0.03	0.87 ± 0.03	0.82 ± 0.02	0.89 ± 0.02
Ensemble	**86.4 ± 1.1%**	0.83 ± 0.02	0.88 ± 0.02	0.86 ± 0.01	0.93 ± 0.01

Table 6.2 – 10-fold stratified cross-validation results (mean ± std across folds)

The cross-validation results are consistent with the held-out test set results, confirming that the performance estimates reported in Chapter 4 are reliable and not artefacts of a favourable test split. The low standard deviations (1.1–1.8% for accuracy, 0.01–0.03 for other metrics) indicate stable generalisation across different partitions of the data.

CHAPTER 7: DISCUSSION

7.1 Interpretation of Results

The empirical results presented in Chapter 4 and Chapter 6 support the central hypothesis of this project: that a hybrid architecture combining structured metric-based prediction with sequential deep learning on raw source code captures complementary predictive signals that yield performance superior to either approach in isolation. This section discusses the implications of the results and situates them within the broader literature.

The 3.7 percentage point improvement of the ensemble over the Random Forest and the 5.4 point AUC-ROC improvement are statistically significant (paired t-test on 10-fold CV fold-level AUC-ROC scores: $t = 6.24$, $p < 0.001$). The magnitude of improvement is modest in absolute terms but meaningful in the context of defect prediction, where the state of the art across published benchmarks has improved by less than 5 percentage points in accuracy over the past decade. These results are therefore consistent with the interpretation that hybrid architectures represent a genuine methodological advance rather than marginal parameter tuning.

The dominance of CBO (Coupling Between Object Classes) as the top SHAP feature is consistent with the theoretical motivation of object-oriented metrics: tightly coupled modules are more likely to be defective because a defect in any dependency may manifest as a failure in the dependent module. The strong showing of process metrics — particularly code churn and number of prior defects — is consistent with the established finding of Moser et al. (2008) that change history is among the most predictive feature categories. Modules that have been heavily modified recently are more likely to contain newly introduced defects, and modules with a history of prior defects are more likely to harbour undiscovered latent defects.

7.2 Comparison with Related Work

The ensemble's AUC-ROC of 0.94 compares favourably with published results on comparable defect prediction tasks. Lessmann et al.'s (2008) comprehensive benchmark found that the best classifiers achieved AUC-ROC values in the range 0.70–0.85 on PROMISE Repository datasets, though direct comparison is complicated by differences in dataset composition and labelling methodology. Dam et al.'s (2018) LSTM-based vulnerability prediction achieved AUC-ROC values of 0.83–0.91 on C/C++ codebases, which is lower than the current system's ensemble AUC-ROC of 0.94 — though this difference may partly reflect the more favourable characteristics of the Python/Java codebase used here.

The most directly comparable published work is that of Li et al. (2017), who combined metric-based features with token-based representations in a two-stage pipeline. Their reported

accuracy of 82.3% is below the current system's 87.1%, though again direct comparison is limited by dataset differences. The current system's advantage likely derives from the more sophisticated LSTM architecture (Li et al. used a simpler bag-of-words token representation rather than a sequential LSTM) and the larger, more diverse training corpus (18,742 modules vs approximately 8,000 in Li et al.'s study).

7.3 Practical Implications

From a practical standpoint, the results suggest that deploying the proposed system in a software development workflow could meaningfully reduce the burden of manual code review. A recall of 0.89 means that the system identifies 89% of all buggy modules, leaving only 11% of defects undetected by automated screening. If developers use the system's predictions to prioritise manual review attention, the probability that a defect escapes both automated detection and the prioritised manual review is substantially lower than with either approach alone.

The precision of 0.84 implies that 16% of modules flagged as buggy are false positives — clean modules incorrectly identified as bug-prone. This false positive rate is substantially lower than the 60%+ rates reported for rule-based static analysis tools (SonarQube, PMD), which are the tools developers encounter in practice. The SHAP-based explanations further mitigate the cost of false positives: even when a module is incorrectly flagged, the associated review suggestion points developers toward genuinely high-complexity or highly-coupled code that, while not defective, may benefit from refactoring.

7.4 Threats to Validity

7.4.1 Internal Validity

The primary internal validity threat is data leakage. Careful steps were taken to prevent leakage: SMOTE was applied only to the training set; normalisation parameters were computed exclusively on the training set; and the validation and test sets remained unchanged throughout all hyperparameter tuning. The use of stratified splitting preserves the original class distribution in all splits, ensuring that evaluation metrics are computed on data with realistic class proportions.

A secondary internal validity concern is the quality of the SZZ-derived labels. SZZ is known to introduce noise through false positives (commits incorrectly classified as bug-fixing) and false negatives (bug-fixing commits not matched by the regex). The meta-commit exclusion heuristic reduces noise but cannot eliminate it entirely. This noise propagates into the training labels, potentially reducing model performance relative to a perfectly labelled dataset.

7.4.2 External Validity

The primary external validity threat is the scope of the evaluation. The dataset consists exclusively of open-source GitHub repositories in Python and Java, selected based on activity

and commit count criteria. This selection may not be representative of: (a) proprietary enterprise codebases, which may have different defect characteristics, coding standards, and team structures; (b) repositories with fewer than 500 commits (typical of small or early-stage projects); or (c) codebases in other programming languages.

The selection criteria favouring active, well-maintained repositories may also introduce a survivorship bias: these repositories may have higher average code quality than the broader population of software projects, potentially making the defect prediction task easier than it would be on lower-quality codebases.

7.4.3 Construct Validity

The primary construct validity concern is the definition of ‘bug-prone’. The SZZ-based labelling equates bug-proneness with having been modified in a bug-fix commit. This definition may not capture all forms of software defect (e.g., performance bugs that do not trigger issue reports, security vulnerabilities discovered through audit rather than failure). The system’s performance on defect types not represented in the training data is unknown.

APPENDIX C: SELECTED CODE LISTINGS

C.1 SZZ Bug-Introducing Commit Identification (Python)

The following listing shows the core SZZ implementation used to identify bug-introducing commits from the repository history:

```
import re from git import Repo  BUG_PATTERN = re.compile(
r'\b(fix|bug|defect|issue|error|patch|resolve|repair|correct)\b',     re.IGNORECASE
)  def identify_buggy_files(repo_path: str) -> dict:     repo = Repo(repo_path)
buggy_files = {}          for commit in repo.iter_commits():          if not
BUG_PATTERN.search(commit.message):              continue          if
len(commit.parents) == 0:              continue                          # Skip meta-
commits (> 50 files changed)          if len(commit.stats.files) > 50:
continue                       for filepath, stats in commit.stats.files.items():
if not filepath.endswith(('.py', '.java')):                  continue             if
filepath not in buggy_files:                 buggy_files[filepath] = []
buggy_files[filepath].append({                  'fix_commit': commit.hexsha,
'lines_changed': stats['lines']              })          return buggy_files
```

C.2 FastAPI Prediction Endpoint (Python)

The following listing shows the /predict/code endpoint implementation:

```
from fastapi import FastAPI, HTTPException from pydantic import BaseModel import
numpy as np  app = FastAPI(title="AI Code Review API", version="1.0.0")  class
CodeRequest(BaseModel):     source_code: str     language: str  # 'python' or
'java'     filename: str = "unnamed"  @app.post("/predict/code") async def
predict_code(request: CodeRequest):     if request.language not in ['python',
'java']:          raise HTTPException(status_code=400,
detail=f"Unsupported language: {request.language}")          # Extract metrics
metrics = extract_metrics(request.source_code, request.language)          #
Tokenise for LSTM     tokens = tokenise(request.source_code, request.language)
tokens_padded = pad_sequence(tokens, max_len=512)          # Inference     rf_prob
= rf_model.predict_proba(          np.array([list(metrics.values())]))[0][1]
lstm_prob = float(lstm_model.predict(          tokens_padded[np.newaxis, :])[0][0])
# Ensemble     ensemble_prob = 0.55 * rf_prob + 0.45 * lstm_prob          # SHAP
explanation     shap_values = shap_explainer.shap_values(
np.array([list(metrics.values())]))[1][0]          return
build_response(ensemble_prob, metrics,                                shap_values,
request.filename)
```

C.3 Angular Code Analysis Service (TypeScript)

```
import { Injectable } from '@angular/core'; import { HttpClient } from
'@angular/common/http'; import { Observable, BehaviorSubject } from 'rxjs'; import
{ tap } from 'rxjs/operators';  @Injectable({ providedIn: 'root' }) export class
AnalysisService {   private apiUrl = environment.apiUrl;   private _loading = new
BehaviorSubject<boolean>(false);   loading$ = this._loading.asObservable();
constructor(private http: HttpClient) {}    analyseCode(code: string, language:
string): Observable<AnalysisResult> {     this._loading.next(true);     return
this.http.post<AnalysisResult>(       `${this.apiUrl}/predict/code`,       {
source_code: code, language }     ).pipe(       tap(() =>
this._loading.next(false))     );   }    analyseMetrics(metrics: Record<string,
number>): Observable<MetricsResult> {     return this.http.post<MetricsResult>(
`${this.apiUrl}/predict/metrics`,       metrics     );   } }
```

APPENDIX D: SAMPLE SYSTEM-GENERATED REVIEW REPORT

The following is a complete sample review report generated by the system for a Python module from the test corpus. All identifying information has been anonymised.

Module	data_processor.py (anonymised)
Language	Python
Analysis Date	2025-09-15 14:32:07 UTC
Ensemble Probability	**0.91 (CRITICAL)**
Detected Smells	God Class (0.87), Long Method (0.79), Feature Envy (0.71)

Key Metrics

Metric	Value	Threshold	Status
CBO	18	10	⚠ **HIGH**
WMC	24	15	⚠ **HIGH**
LOC	847	300	⚠ **HIGH**
Cyclomatic Complexity	9.4	7	⚠ **ELEVATED**
LCOM	0.81	0.5	⚠ **HIGH**
Code Churn (6mo)	1,247	500	⚠ **HIGH**
DIT	2	5	✓ OK
NOC	0	N/A	✓ OK

Review Suggestion

CRITICAL: This module exhibits multiple high-severity quality issues. CBO = 18 (threshold: 10) indicates critically high inter-module coupling; this module depends on 18 other classes, making it fragile to changes in any of its dependencies. WMC = 24 (threshold: 15) indicates excessive total method complexity; the average cyclomatic complexity per method of 9.4 exceeds the recommended maximum of 7, suggesting methods contain complex branching logic that should be decomposed. The module has accumulated 1,247 lines of code churn in the past 6 months, making it a high-risk target for newly introduced defects. Recommended actions: (1) Apply the Single Responsibility Principle — decompose this class into 3-4 smaller classes each with a clearly defined responsibility; (2) Reduce inter-module coupling through dependency inversion — introduce interfaces to decouple from concrete implementations; (3) Refactor methods with cyclomatic complexity above 10 into smaller, testable units;

(4) Increase test coverage for this module given its high churn rate. Estimated refactoring effort: 2-3 developer days.

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APPENDIX A: SOFTWARE DEPENDENCIES AND VERSIONS

Package / Tool	Version	Purpose
Python	3.10.12	Primary programming language
TensorFlow / Keras	2.12.0	LSTM model training and inference
Scikit-learn	1.3.0	Random Forest, preprocessing, evaluation
SHAP	0.42.1	Model explainability and feature attribution
FastAPI	0.103.1	REST API backend framework
Uvicorn	0.23.2	ASGI server for FastAPI
Angular	16.2.0	Frontend web dashboard
Radon	5.1.0	Python static code metrics
CKJMExtended	2.3	Java object-oriented metrics
Pygments	2.15.1	Source code tokenisation
imbalanced-learn	0.11.0	SMOTE class balancing
PyGitHub	1.59.1	GitHub REST API client
joblib	1.3.2	Random Forest serialisation
NumPy	1.24.3	Numerical computing
Pandas	2.0.3	Data manipulation
Matplotlib	3.7.2	Visualisation (training curves)
CUDA	11.8	GPU acceleration for LSTM
cuDNN	8.6.0	Deep learning GPU primitives
Ubuntu	22.04 LTS	Operating system
Node.js	18.17.0	Angular frontend build

APPENDIX B: GITHUB REPOSITORY SAMPLE

The following table lists a representative sample of 20 GitHub repositories from the 512 used in the training corpus, illustrating the diversity of projects, languages, sizes, and domains included.

#	Repository (anonymised)	Language	Commits	Domain
1	org-a/web-framework	Python	4,821	Web development
2	org-b/data-pipeline	Python	2,103	Data engineering
3	org-c/ml-library	Python	8,472	Machine learning
4	org-d/network-tools	Python	1,587	Networking
5	org-e/test-framework	Python	3,214	Testing
6	org-f/cli-toolkit	Python	976	Developer tools
7	org-g/enterprise-app	Java	11,284	Enterprise software
8	org-h/android-sdk	Java	6,731	Mobile development
9	org-i/microservices	Java	2,845	Distributed systems
10	org-j/spring-extension	Java	1,962	Web framework
11	org-k/database-driver	Java	3,518	Database
12	org-l/security-lib	Java	1,284	Security
13	org-m/compiler-tools	Python	2,967	Compilers
14	org-n/api-gateway	Java	4,103	API management
15	org-o/data-viz	Python	1,847	Visualisation
16	org-p/orm-library	Java	5,621	Database
17	org-q/devops-tools	Python	3,102	DevOps
18	org-r/cloud-client	Java	2,784	Cloud computing
19	org-s/nlp-toolkit	Python	4,218	NLP
20	org-t/monitoring	Java	1,563	Observability

APPENDIX E: PROJECT MANAGEMENT AND TIMELINE

E.1 Project Schedule

The project was executed over two semesters (Fall 2024 and Spring 2025) across eight phases. The following Gantt-style summary table describes the planned and actual timelines for each phase.

Phase	Planned Start	Planned End	Duration	Status
Literature Review & Problem Definition	Sep 2024	Oct 2024	6 weeks	**Completed**
Data Collection Pipeline	Oct 2024	Nov 2024	5 weeks	**Completed**
Feature Engineering & Dataset Construction	Nov 2024	Dec 2024	4 weeks	**Completed**
Random Forest Model Development	Dec 2024	Jan 2025	3 weeks	**Completed**
LSTM Model Development	Jan 2025	Feb 2025	5 weeks	**Completed**
Ensemble Integration & Evaluation	Feb 2025	Mar 2025	3 weeks	**Completed**
API & Frontend Development	Mar 2025	Apr 2025	5 weeks	**Completed**
Testing, Documentation & Report Writing	Apr 2025	Sep 2025	5 months	**Completed**

Table E.1 – Project timeline summary

E.2 Team Responsibilities

The project was executed by three team members, with responsibilities divided as follows:

Team Member	Primary Responsibilities
Umair Dost	LSTM model architecture and training; FastAPI backend development; SHAP explainability integration; ensemble integration; API testing; report writing (Chapters 3, 4)
Muhammad Nasir Khan	GitHub data collection pipeline; SZZ algorithm implementation; feature engineering and selection; Random Forest model development; cross-validation; report writing (Chapters 1, 2)
Ahmed Khan Khisro	Angular frontend development; UI/UX design; system integration testing; load testing; report writing (Chapters 5, 6, 7); documentation

E.3 Tools and Infrastructure

Development was performed using the following tools and infrastructure:

- Version Control: Git with GitHub (private repository); branching strategy: feature branches merged to main via pull requests with peer code review.
- Development Environment: VS Code (Python backend), WebStorm (Angular frontend), Jupyter Notebooks (exploratory data analysis and model prototyping).
- Experiment Tracking: MLflow (version 2.6) for tracking model training runs, hyperparameter configurations, and metric histories.
- Containerisation: Docker and Docker Compose for reproducible development environments and deployment packaging.
- Continuous Integration: GitHub Actions workflow running unit and integration tests on every push to the main branch.

APPENDIX F: ETHICAL CONSIDERATIONS

F.1 Data Ethics

All data used in this project was sourced from publicly accessible GitHub repositories under open-source licences. Repository selection was limited to repositories explicitly made public by their authors, and no private or proprietary code was accessed or included in the dataset. The use of public GitHub repositories for machine learning research is consistent with established practices in the Mining Software Repositories community and is permitted under GitHub's Terms of Service for non-commercial research purposes.

Repository and author identifying information was not used as a predictive feature and does not appear in the trained models. The anonymised repository identifiers used in Appendix B are randomly generated labels that cannot be reverse-mapped to actual repositories without access to the original data collection logs, which are retained securely by the project team.

F.2 Responsible AI Considerations

The system is designed to augment developer judgment, not to replace it. Prediction outputs are clearly labelled as probabilistic assessments rather than deterministic verdicts, and the SHAP-based explanations are designed to provide developers with the information needed to critically evaluate each prediction rather than blindly accept it. The review suggestions are framed as recommendations rather than mandates.

The system does not make decisions about code acceptance or rejection autonomously; it provides information to support human decision-making. Developers retain full agency over whether to act on the system's recommendations. This design choice reflects an ethical commitment to keeping humans in the loop for consequential quality assurance decisions.

F.3 Bias and Fairness

The training corpus is skewed toward active, well-maintained open-source repositories on GitHub, which may not be representative of all software development contexts. The system's performance on codebases from underrepresented languages, domains, or development practices has not been evaluated. Practitioners deploying the system on codebases with characteristics substantially different from the training corpus should validate performance on a representative sample of their own data before relying on the system's predictions.

Metric-based features such as code churn and number of authors are influenced by team size and project age, which are correlated with organisational factors (project funding, team experience). The system does not account for these confounders, and its predictions should not

be used to make comparative quality assessments across projects with substantially different resource levels or development histories.